Graph Paths I

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1723 | Mathematics | https://cses.fi/problemset/task/1723

Consider a directed graph that has n nodes and m edges. Your task is to count the number of paths from node 1 to node n with exactly k edges.

Input

The first input line contains three integers n, m and k: the number of nodes and edges, and the length of the path. The nodes are numbered 1,2,…,n.

Then, there are m lines describing the edges. Each line contains two integers a and b: there is an edge from node a to node b.

Output

Print the number of paths modulo 10^9+7.

Constraints

- $1 \le n \le 100$
- $1 \le m \le n(n-1)$
- $1 \le k \le 10^9$
- $1 \le a,b \le n$

Example

Input
3 4 8 1 2 2 3 3 1 3 2

Output
2

Explanation: The paths are 1 → 2 → 3 → 1 → 2 → 3 → 1 → 2 → 3 and 1 → 2 → 3 → 2 → 3 → 2 → 3 → 2 → 3.